
Bilateral AI: Agentic AI Systems as Neuro-Symbolic Integration

Karolina Drożdż¹ Kristian Kersting² Günter Klambauer³ Robert Legenstein⁴
Mena Leemhuis³ Francesca Toni⁵

¹IDEAS Warsaw & ELLIS Unit Warsaw ²TU Darmstadt & ELLIS Unit Darmstadt ³JKU Linz & ELLIS Unit Linz

⁴TU Graz & ELLIS Unit Graz ⁵Imperial College London

Proposal for a NeurIPS 2026 workshop, Dec 12/13, Paris, Europe.

Tagline. Every LLM agent is a neuro-symbolic system: a neural network inside a symbolic harness. Bridging three communities that need each other.

Logistics. Location preference: Paris only. **Workshop website:** <https://ml-jku.github.io/bilateral-ai/>. **Concurrent submission disclosure:** all but one organizer (Günter Klambauer, see below) are involved only in this proposal. **Estimated number of attendees:** 200. **Estimated number of papers:** ~30. **Past iterations / held before:** none, but the organizers have substantial experience organizing workshops. **Special requirements and technical needs:** A/V; panel. **We commit to reviewing on OpenReview. The workshop is non-archival.**

Description of Workshop Topic and Content

Modern agentic AI systems consist of a sub-symbolic large language model paired with a symbolic harness that parses outputs, dispatches tools, maintains state, and enforces control flow. Every deployed agent is therefore a *bilateral AI* system, structurally indistinguishable from the integrated architectures the neuro-symbolic community has studied for decades (Mao et al., 2025; Colelough and Regli, 2025). **Where agents fail.** Current systems lose their goals across long tool-call sequences, fail to recognise task completion, coordinate poorly, and fail surprisingly on simple problems. AgentBench identifies poor long-term reasoning and instruction-following as dominant obstacles (Liu et al., 2024). MIRAGE-Bench formalises agent hallucinations as actions unfaithful to task instructions or environment observations, including fabricated tool calls (Zhang et al., 2025). Agent-SafetyBench finds that none of sixteen evaluated agents reaches a 60% safety score, attributing the gap to structural defects that prompting cannot fix (Zhang et al., 2024). **Why neuro-symbolic AI holds the answers.** These failures sit in bookkeeping, compositional planning, and self-verification rather than perception. Decades of neuro-symbolic work on compositional generalisation, state-tracking, and hierarchical memory bear directly on them (Mao et al., 2025). The workshop frames agentic AI as neuro-symbolic systems, treats the symbolic harness as a research object rather than infrastructure, and brings together three communities that currently meet only by accident: (a) neuro-symbolic AI and knowledge representation, (b) LLM agents and tool use, and (c) benchmarking of agentic AI systems. Workshop details below.

Invited speakers

Sepp Hochreiter (Johannes Kepler University Linz, ELLIS Unit Linz & Institute for Machine Learning; [website](https://www.ml.jku.at), GS.hochreit@ml.jku.at; **CONFIRMED**¹). Sepp Hochreiter heads the Institute for Machine Learning and the Linz Institute of Technology AI Lab at Johannes Kepler University Linz (JKU). He is the

¹Minor uncertainty about joining NeurIPS; would give a virtual talk otherwise



research director of Austria’s AI Cluster of Excellence and director of the ELIS Unit Linz, chief scientist at NXAI and a founding director of the Institute of Advanced Research in Artificial Intelligence. He is a pioneer of deep learning and inventor of the long short-term memory (LSTM) neural network architecture, and has also contributed to meta-learning and reinforcement learning. He and his team recently built large language models based on the xLSTM architecture.



Marta Kwiatkowska (*University of Oxford; [website](#), [GS](#), Marta.Kwiatkowska@cs.ox.ac.uk; **CONFIRMED**¹). Marta Kwiatkowska is Professor of Computing Systems and Fellow of Trinity College at the University of Oxford. She spearheaded probabilistic and quantitative methods in verification and now works on safety and robustness for machine learning and AI. She led the development of the PRISM model checker, the leading tool in the area, and is the first female winner of the 2018 Royal Society Milner Award. Her work on formal verification and guarantees for complex systems*

speaks directly to the workshop’s central question: what the symbolic harness should guarantee and how those properties can be established.



Mario Zechner (*Earendil & Independent; [website](#), [GS](#), contact@mariozechner.at; **Tentative Yes**). Mario is the author of Pi, an opinionated and minimal coding agent, and writes extensively on the design of agent harnesses: tool dispatch, the trade-offs of MCP versus CLI interfaces for coding agents, and the principle that “prompts are code and state lives in files.” Pi is also the basis of OpenClaw. His hands-on, systems-level perspective on how an LLM is wired into a symbolic control layer brings a practitioner’s voice on the harness-as-research-object theme at the heart of this workshop.*



Michela Milano (*Università di Bologna; [website](#), [GS](#), michela.milano@unibo.it; **CONFIRMED**¹). Michela Milano is Full Professor of Artificial Intelligence at the Department of Computer Science and Engineering (DISI), University of Bologna. Her research focuses on decision support, combinatorial optimization, and the integration of machine learning with constraint programming. She is Editor-in-Chief of the *Constraints* journal, past Deputy President of EurAI, and past Executive Councilor of AAAI. In 2016 she received the Google Faculty Research Award. Her work*

integrating learning with constraint-based reasoning is a direct instance of the workshop’s theme: coupling a sub-symbolic learner to a symbolic component that enforces structure and constraints.



Petar Veličković (*Google DeepMind; [website](#), [GS](#), petarv@google.com; **CONFIRMED**¹). Petar Veličković is a senior staff research scientist at Google DeepMind and affiliated lecturer at the University of Cambridge, where he earned his PhD. His research aligns neural networks to classical computation to improve out-of-distribution generalisation, with a focus on neural algorithmic reasoning, graph representation learning, and geometric deep learning. He is widely recognised for his foundational work on Graph Neural Networks. His research on aligning neural networks with algorithmic computation goes to*

the heart of the workshop’s question: how a sub-symbolic learner can be coupled with symbolic, rule-governed computation.



Marco Gori (*University of Siena; [website](#), [GS](#), marco.gori@unisi.it; **CONFIRMED**¹). Marco Gori is Full Professor of Computer Science at the University of Siena, where he leads the Siena Artificial Intelligence Lab. His research focuses on machine learning and neural computation. He is widely recognised as a pioneer of Graph Neural Networks, having co-coined the term in the IJCNN 2005 paper “A New Model for Learning in Graph Domains” and consolidated the framework in a 2009 IEEE-TNN paper now exceeding 14,000 citations. He is a Fellow of IEEE, EurAI, IAPR, and ELLIS. His work on learning over structured, relational representations connects neural computation to the*

symbolic structures at the core of the workshop’s neuro-symbolic focus.

Barbara Hammer (*Bielefeld University; [website](#), [GS](#), bhammer@techfak.uni-bielefeld.de; **CONFIRMED**¹). Barbara Hammer is Professor of Machine Learning at the Center for Cognitive Interaction Technology (CITEC), Bielefeld University, and elected DFG representative for Machine Learning.*



Her expertise spans intelligent data analysis, explainable machine learning, and trustworthy AI, with current work on learning for non-Euclidean and graph-structured data, model explainability, and online learning under drift. Her work on explainability and learning with drift bears directly on the workshop’s concern with agents that must track state and remain interpretable as conditions change over long trajectories.

Diversity statement. We have a highly diverse set of speakers. Our six invited speakers are **gender-balanced**, span four European countries (Austria, UK, Germany, Italy), and **represent academia, industry** (Veličković/Google DeepMind), and a **university spinoff-startup** (Hochreiter/NXAI). Career stages range from senior research scientist to full professor. The European focus is intentional given our Paris location, reducing travel barriers while drawing on a region with exceptional depth in this area. Our call for contributions will explicitly encourage submissions from early-career and underrepresented researchers, and we will seek travel grants where funding allows.

Workshop details

Differentiation to other workshops. NeurIPS 2026 will host many workshops on agentic AI, LLM reasoning, and benchmarking. They take the symbolic harness for granted and ask what the model should do inside it. **We invert this: the harness is a research object.** We ask what it is formally, what it should guarantee, and how its design determines where agents fail. This reframing is what lets us convene three communities that currently meet only by accident: (a) neuro-symbolic AI and knowledge representation, (b) LLM agents and tool use, and (c) the LLM and agentic AI benchmarking community. Agent-builders construct harnesses daily without a theory of them; neuro-symbolic researchers have studied exactly these problems for decades under different names; benchmarking grounds the question empirically. Benchmarking and LLM contributions are welcome, but in service of one question no existing workshop is asking: **what is the interplay between the neural component (the LLM) and the symbolic component (the harness), and how does it affect properties of the resulting agent?**

Why now? Agentic AI is being deployed at scale while its symbolic harness is still built ad hoc, and the three communities that should jointly address its failures currently work in isolation. Without deliberate coordination now, the field risks reinventing decades-old neuro-symbolic results under new names and entrenching fragmented, non-comparable benchmarks; convening these groups this year is the moment to align vocabulary, problems, and evaluation before that divergence hardens.

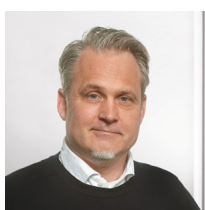
Timeline. CfP mid-July; submissions due Aug 21; reviewing Aug 22–Sep 18 (3 reviews/paper, ≤ 3 papers/reviewer, via OpenReview); author notification Sep 22 (before the mandatory Sep 29 deadline). Contributions are non-archival; titles posted before site publication. **Schedule.** Full-day, three thematic blocks (one per community). Each block: 2–3 invited talks, 1 contributed talk, extensive moderated discussion (**$\sim 40\%$ discussion time; $\sim 60\%$ talks**). Plus a cross-community panel (“What should the symbolic harness guarantee?”) and an end-of-day poster session. Organizers introduce/moderate only; contributed slots prioritize early-career authors. The workshop is intended to be a large-attendance talk format. **COI.** All papers are decided by the Program Committee, not organizers. Each submission is handled by a subcommittee of conflict-free PC members (multiple subcommittees if needed); no one assesses same-organization work. Organizers and their students/postdocs may not submit; organizer-linked submissions are routed to subcommittees excluding the conflicted parties. **Diversity efforts.** The CfP explicitly encourages early-career and underrepresented authors; contributed-talk slots are reserved for early-career first authors; travel subsidies are sought where funding allows; the CfP is circulated beyond our networks; and the PC is recruited across geography, career stage, and academia/industry. **Prior publication policy.** We will state in the CfP that the workshop is for work in progress and unpublished results; work already published at NeurIPS or other ML/AI venues (including main-conference papers and invited-talk content) is not eligible, and we will screen submissions for this at review. **Public workshop information.** At the CfP stage, the workshop page will list the invited speakers with preliminary talk titles, the timeline, and clearly state that submissions are non-archival. Titles of accepted papers and the full schedule will be added by the end of November. The organizers will also ensure that the schedule, including titles, is available in the conference app.

Organizer information



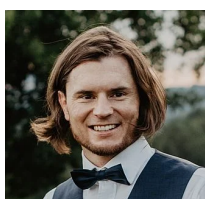
Karolina Drożdż (*IDEAS Warsaw & ELLIS Unit Warsaw; [website, GS, karolina.drozdz@ideas.edu.pl](https://www.guest.ac.uk/people/karolina.drozdz@ideas.edu.pl)*). Karolina Drożdż is a human-centric AI researcher at IDEAS and a doctoral student affiliated with the ELLIS Unit Warsaw, holding a Research Master in Neuroscience and AI from the University of Amsterdam. Her work bridges cognitive science, psychology, and deep learning to study LLMs’ functional capacities for introspection, metacognition, and situation modelling, with a focus on interpretability and trustworthiness. She is active in interdisciplinary community building, coordinating neuroscience–

CS projects and organising seminars and scientific events. Her cross-community perspective and cognitively grounded evaluation methods make her an excellent fit to facilitate dialogue across the workshop’s three communities and to extend its reach into the growing AI community in Central and Eastern Europe.



Kristian Kersting (*TU Darmstadt & ELLIS Unit Darmstadt; [website, GS, kristian.kersting@tu-darmstadt.de](https://www.guest.ac.uk/people/kristian.kersting@tu-darmstadt.de)*). Kristian Kersting is a Full Professor in the Computer Science Department at TU Darmstadt. He heads the Artificial Intelligence and Machine Learning (AIML) lab, is a member of the Centre for Cognitive Science, a faculty of the ELLIS Unit Darmstadt, and the founding co-director of the Hessian Center for Artificial Intelligence (hessian.AI). He is head of the Research Department on the Foundations of Systems AI at the German Research Center for Artificial Intelligence (DFKI) and a Fellow of the Association for the Advancement of Artificial Intelligence (AAAI). His main

research interests are statistical relational AI, neuro-symbolic AI, and deep (probabilistic) programming and learning. As Director of the ELLIS program on “Semantic, Symbolic and Interpretable Machine Learning” he regularly organizes events on similar topics.



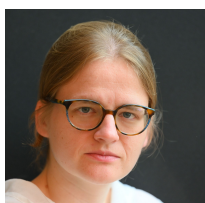
Günter Klambauer (*Johannes Kepler University Linz & ELLIS Unit Linz; [website, GS, klambauer-office@ai-lab.jku.at](https://www.guest.ac.uk/people/guenther.klambauer-office@ai-lab.jku.at)*). Günter Klambauer is a Full Professor for Artificial Intelligence in Life Sciences at the Institute for Machine Learning at Johannes Kepler University Linz, Austria. He is a key member of the LIT AI Lab, a leading hub for AI research in Austria. He is a Scholar in ELLIS, and led the “Machine Learning for Molecule Discovery” programme as ELLIS Director from December 2023 until 2025. He is known for the development of self-normalizing neural networks. Key areas of his research

include deep learning and architectures, signal propagation theory, and machine learning methods for life sciences. He led the Machine Learning for Molecules Workshop series from 2021–2025 with up to 700 participants. GK is involved in one other proposal.



Robert Legenstein (*Graz University of Technology & ELLIS Unit Graz; [website, GS, robert.legenstein@tugraz.at](https://www.guest.ac.uk/people/robert.legenstein@tugraz.at)*). Robert Legenstein is Full Professor and head of the Institute of Machine Learning and Neural Computation at Graz University of Technology (TU Graz). He is a founding member of the European AI Excellence Initiative ELLIS and ELLIS Fellow at the ELLIS Unit Graz. He played a key role in founding the Graz Center for Machine Learning (GraML). He is a computer scientist working in computational/theoretical neuroscience, novel brain-inspired computing paradigms, neural networks, and neuro-symbolic AI. As head of GraML he organized large scientific and

public events (GraML conference, GraML industry day) with ~100 participants, and he co-organized workshops on spiking neural networks at the AIROV conference.



Mena Leemhuis (*Johannes Kepler University Linz; [website, GS, mena.leemhuis@jku.at](https://www.guest.ac.uk/people/mena.leemhuis@jku.at)*). Mena Leemhuis is a tenure-track professor for Neuro-Symbolic AI at the Institute for Symbolic AI at Johannes Kepler University Linz. Before, she was a postdoctoral researcher at the KRDB research centre for Knowledge-based Artificial Intelligence at the Free University of Bozen-Bolzano. Her work focuses on the tight integration of neural and symbolic methods in the context of knowledge graphs and ontologies, with a particular focus on interpretability and trustworthiness of such methods. She is active in

the community, e.g., as speaker at the Austrian Computer Science Day, lecturer at the ESSAI summer school, and co-organizer of the doctoral consortium at the KI conference.



Francesca Toni (*Imperial College London*; [website](#), [GS](#), f.toni@imperial.ac.uk). Francesca Toni is a Professor in Computational Logic in the Department of Computing at Imperial College London, the leader of the CLArg (Computational Logic and Argumentation) research group, and the founder and co-director of the Centre for eXplainable AI (XAI). Her research interests lie within the broad area of Explainable Artificial Intelligence with a current focus on machine arguing and computational argumentation, including argumentation in AI broadly conceived and argument

mining from text. She is IJCAI-ECAI 2026 general chair and has been programme chair of KR 2018 and COMMA 2022, amongst others. She is also a keynote speaker at NeSy 2026. She enriches the organizer team with expertise in explainable AI, neuro-symbolic learning, and argumentative reasoning with LLMs and beyond.

Diversity statement. Our six-person organizing team is highly diverse across multiple axes. It is gender-balanced (3 women, 3 men) and spans career stages from doctoral student (Drożdż) through tenure-track professor (Leemhuis) to full professor (Kersting, Klambauer, Legenstein, Toni). Geographically it brings together four countries and institutions across Europe: Poland (IDEAS/ELLIS Warsaw), the UK (Imperial College London), Germany (TU Darmstadt), and Austria (JKU Linz, TU Graz), deliberately extending beyond the Western-European core into Central and Eastern Europe. The team also bridges complementary research traditions (neuro-symbolic AI, computational logic and argumentation, cognitive science, and machine learning theory), reflecting the cross-community spirit of the workshop itself.

Program committee

The following **30 researchers** have already agreed that they would join the program committee, and we anticipate around 60 members to cover the review easily: Devendra Singh Dhami; Thomas Eiter; Fajar J. Ekaputra; Alexander Felfernig; Johannes Fürnkranz; Bernhard Geiger; Dietmar Jannach; Roman Kern; Umair Khan; Erich Kobler; Simon Kohaut; Bettina Könighofer; Nada Lavrac; Robert Legenstein; Xinghan Liu; Wolfgang Maass; Marc Masana; Saurabh Mathur; Marco Mondelli; Michael Morak; Nysret Musliu; Satya Prakash Nayak; Ozan Özdenizci; Axel Polleres; Van Quoc Phuong Huynh; Diego Rincon-Yanez; Marta Sabou; Emanuel Sallinger; Olga Saukh; Konstantin Schekotihin.

References

- Colelough, B. C. and Regli, W. (2025). Neuro-symbolic ai in 2024: A systematic review. *arXiv preprint arXiv:2501.05435*.
- Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Yang, K., et al. (2024). Agentbench: Evaluating llms as agents. In *International Conference on Learning Representations*, volume 2024, pages 52989–53046.
- Mao, J., Tenenbaum, J. B., and Wu, J. (2025). Neuro-symbolic concepts. *arXiv preprint arXiv:2505.06191*.
- Zhang, W., Sun, Y., Huang, P., Pu, J., Lin, H., and Song, D. (2025). Mirage-bench: Llm agent is hallucinating and where to find them. *arXiv preprint arXiv:2507.21017*.
- Zhang, Z., Cui, S., Lu, Y., Zhou, J., Yang, J., Wang, H., and Huang, M. (2024). Agent-safetybench: Evaluating the safety of llm agents. *arXiv preprint arXiv:2412.14470*.